

Graphical abstract — caption

Article: *Ambient WiFi as a Cheap Radar for Automotive SLAM: Centimetre Localization, a Phantom Ceiling on Mapping, and What That Ceiling Is — in Simulation and on \$30 of Silicon*

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Ambient WiFi on a moving vehicle localizes to 0.098 m — a statistical tie with LiDAR at 84–600× lower sensor cost — yet its map scores zero: $\approx 89\%$ of realistic-CSI detections are phantoms (controlled scene). Ablation isolates the cause as geometry, not carrier, and wider bandwidth worsens it. Reproduced on two \$30 ESP32-S3 boards.